# Jai Soni

ms23btech11011@iith.ac.in | sonijai737@gmail.com | +91 9116667956 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 7.77 |
| Tagore Public School Jaipur | Class XII (Central Board of Secondary Education) | 2022 | 87.20% |
| St pauls Sr Sec School Jaipur | Class X (Central Board of Secondary Education) | 2020 | 88.20% |

## EXPERIENCE

**ServiceNow**

Machine Learning Intern (Infrastructure & Platform)

- **Migrated legacy CI/CD pipelines** to a unified **Go-based workflow** for ServiceNow's internal ML platform, retiring legacy infrastructure and **reducing monthly compute spend by an estimated 30%.**
- Diagnosed a **Docker Hub pull-rate-limit** bottleneck that was halting CI/CD merges across two externally-hosted clusters outside ServiceNow's network, and resolved it by **migrating image pulls to ServiceNow's internal, rate-limit-free registry** restoring uninterrupted merge capability for the full engineering team.
- Re-architected a brittle batch artifact-copy process into a **fault-tolerant** pipeline that logs and bypasses failed items instead of halting entirely, boosting pipeline reliability and **minimizing manual intervention**
- Diagnosed a critical **networking** bug causing interactive terminal sessions to silently freeze for up to 15 minutes, and resolved it with a **keepalive heartbeat** mechanism that detects and recovers dead connections.
- Shipped a **real-time CLI job-status streaming** feature enabling autonomous, human-free pipeline chaining, plus a **full-text log search** tool in the web dashboard, sharply improving debuggability for ML researchers platform-wide.
- Engineered a near-instant **dataset deletion mechanism** by implementing asynchronous background folder isolation, entirely eliminating UI wait times for researchers managing massive ML training sets.
- Hardened system stability by building automated **permission auto-correction** for directory crashes and expanding platform capability to parse complex, multi-line configuration values like security certificates.

## PROJECTS

**Distributed Rate Limiter Service**

- Engineered a distributed rate-limiting microservice in **Go** with a **REST API** for runtime policy configuration, implementing **Token Bucket** and **Sliding Window** algorithms for per-client request throttling.
- Utilized **Redis** with custom **Lua scripts** for atomic counter execution, eliminating race conditions across high-concurrency requests without the need for service restarts.
- Orchestrated multi-stage **Docker** containers on **Kubernetes** with HPA and health probes; integrated **Prometheus** metrics for real-time traffic observability.

**Smart Hiring Assistant**

- Architected an AI-powered **Applicant Tracking System** leveraging **Sentence-BERT** to rank candidate profiles using a weighted scoring model (60% semantic similarity, 30% skill alignment, 10% education).
- Engineered a custom **NLP extraction pipeline** with keyword expansion and token filtering to accurately match technical skills against complex job requirements.
- Deployed a scalable **Flask backend** with a responsive UI supporting automated, multi-format resume parsing (PDF, DOCX, TXT) for real-time candidate evaluation.

**AskCampus- Context-Aware RAG Assistant**

- Engineered a **Retrieval-Augmented Generation (RAG)** data pipeline using **BeautifulSoup** to scrape institutional data and curated FAQs, indexing them via **FAISS vector search** and **OpenAI embeddings.**
- Integrated **LangChain** workflows behind a **Flask** web backend to deliver accurate, context-grounded answers on

academics and campus procedures.
- Developed a responsive frontend interface to seamlessly execute similarity searches over vector indexes with minimal retrieval latency.

## ACHIEVEMENTS

- **Synchrony Hackathon**: Secured a spot among the **Top 10 teams** out of **50 participating teams.**
- **Competitive Programming**: Solved 600+ problems on **LeetCode** and 300+ problems on **Codeforces**.
- **Internship Coordinator (OCS)**: Facilitated corporate outreach and placement logistics for the Office of Career Services at IIT Hyderabad.

## TECHNICAL SKILLS

**Languages**: C, C++, Python, Golang, Java

**Frameworks & Libraries**: Flask, LangChain, FAISS

**Backend & Infrastructure**: Redis, REST APIs, Docker, Kubernetes

**Core CS**: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems **Developer Tools**: Linux, Jenkins, Git, GitHub, CI/CD

## RELEVANT COURSEWORK AND CERTIFICATIONS

**Computer Science Courses**: Introduction to Programming, Introduction to Artificial Intelligence, Fundamentals of Scientific Computing, Computational Methods

**Mathematics Courses**: Calculus I, Calculus II, Linear Algebra, Differential Equations

**Certifications**: Machine Learning Specialization by Andrew Ng (Coursera), Deep Learning Specialization by DeepLearning.AI, Natural Language Processing by DeepLearning.AI, Advanced Computer Vision with TensorFlow (Coursera)